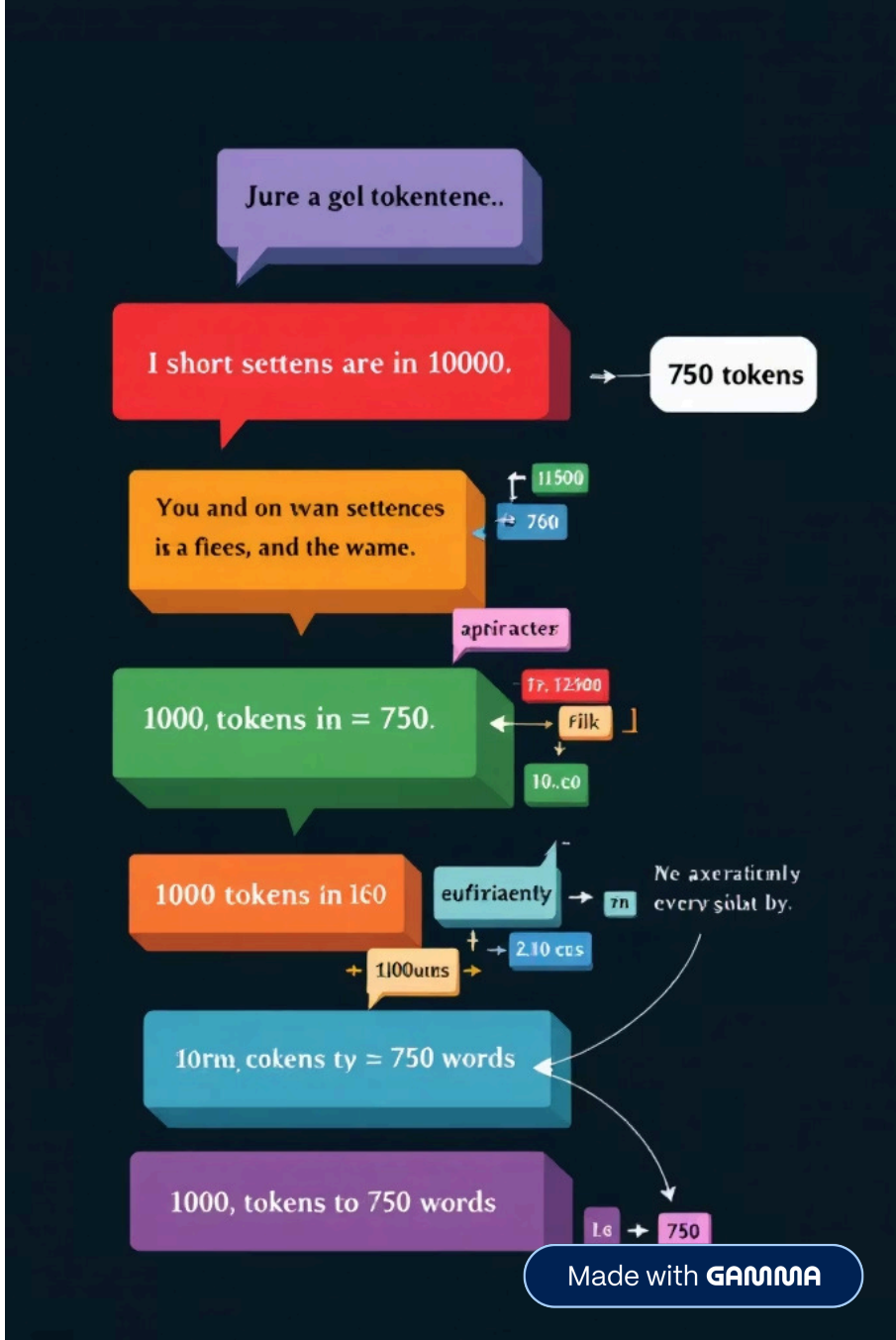
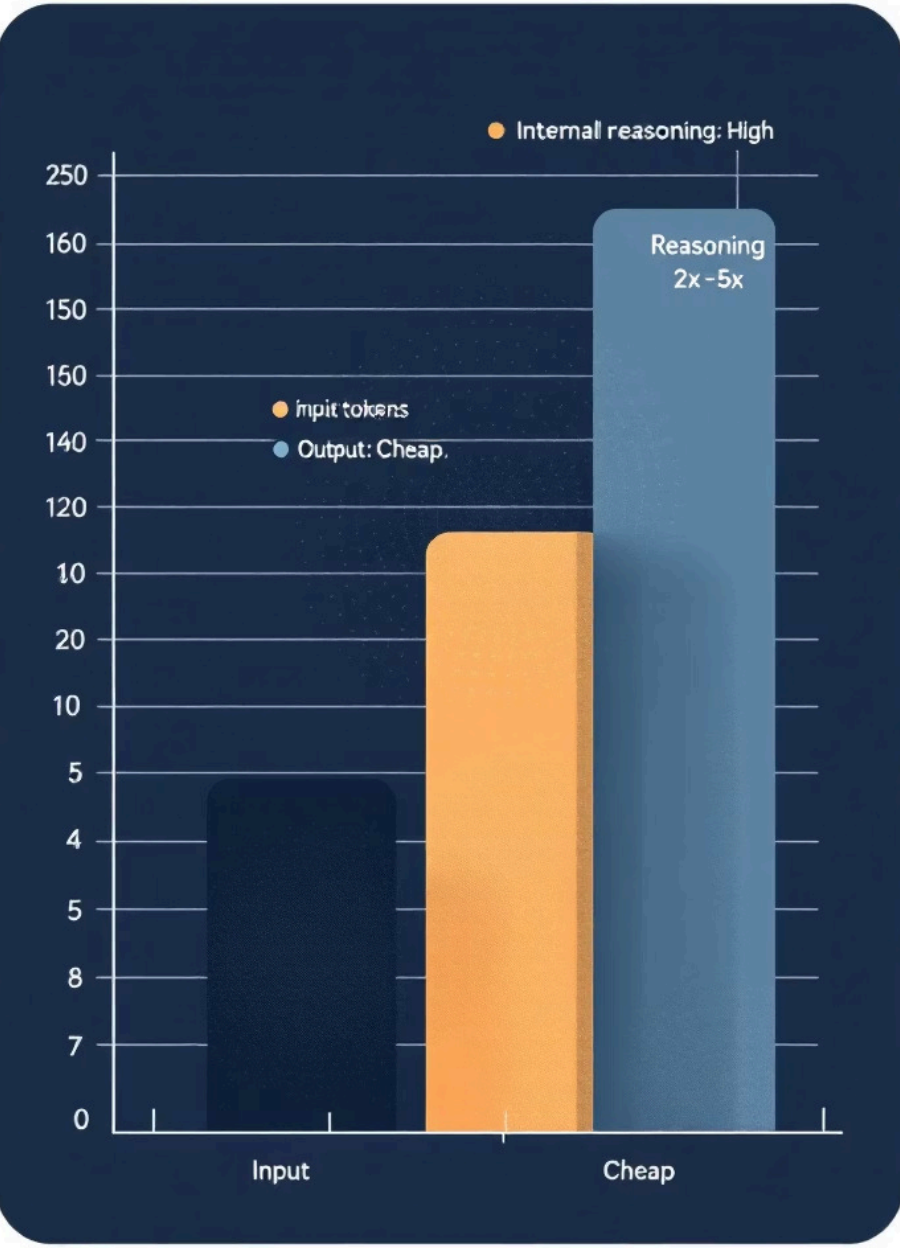
What Is a Token?

A token is the fundamental unit that LLMs use to process text. Rather than reading character by character or word by word, language models break text into smaller chunks called tokens.

General rules: 1,000 tokens ≈ 750 words in English; one token ≈ 4 characters; a typical page (500 words) ≈ 666 tokens.

Important: tokenization varies by model and provider — the same sentence can yield different token counts depending on the tokenizer.





Token Types & Pricing

- ## Input Tokens

  Text you send in requests including system prompts, user messages, context documents, conversation history, and few-shot examples. Generally less expensive than output tokens because the model only processes and understands them.

- ## Output Tokens

  Text generated by the model in response to your input. Usually cost 2-5x more than input tokens. Each output token requires the model to calculate probability distributions across its entire vocabulary.

- ## Thinking Tokens

  Internal reasoning process in models like OpenAI's o3 and Claude with extended thinking. May generate 10-30x more tokens than visible output, significantly impacting costs for complex reasoning tasks.

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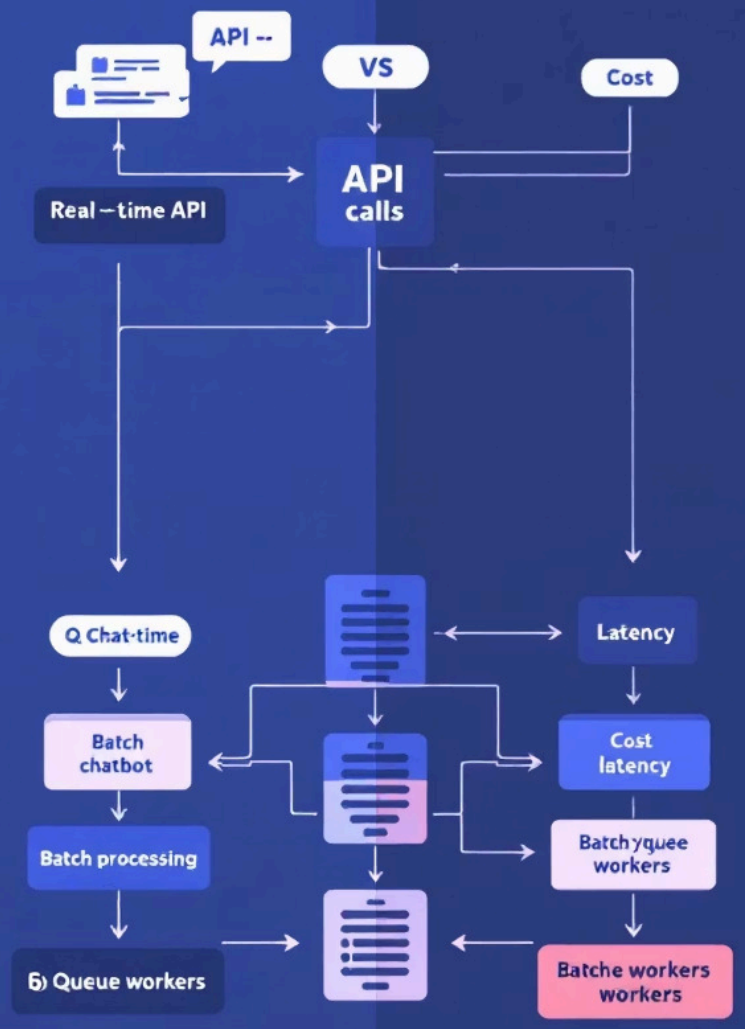

API Request Types

Standard API Calls

Synchronous requests where you send a prompt and receive immediate response. Ideal for real-time applications, interactive chatbots, user-facing features where latency matters, and streaming responses. Priced at full listed rate with higher rate limits.

Batch API Processing

Asynchronous bulk jobs with results typically returned within 24 hours. Most providers offer 50% discount on both input and output tokens. Ideal for large-scale data processing, document summarization, content generation, evaluation pipelines, and any workflow where immediate results aren't required.

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Prompt Caching to Save Costs

How it works

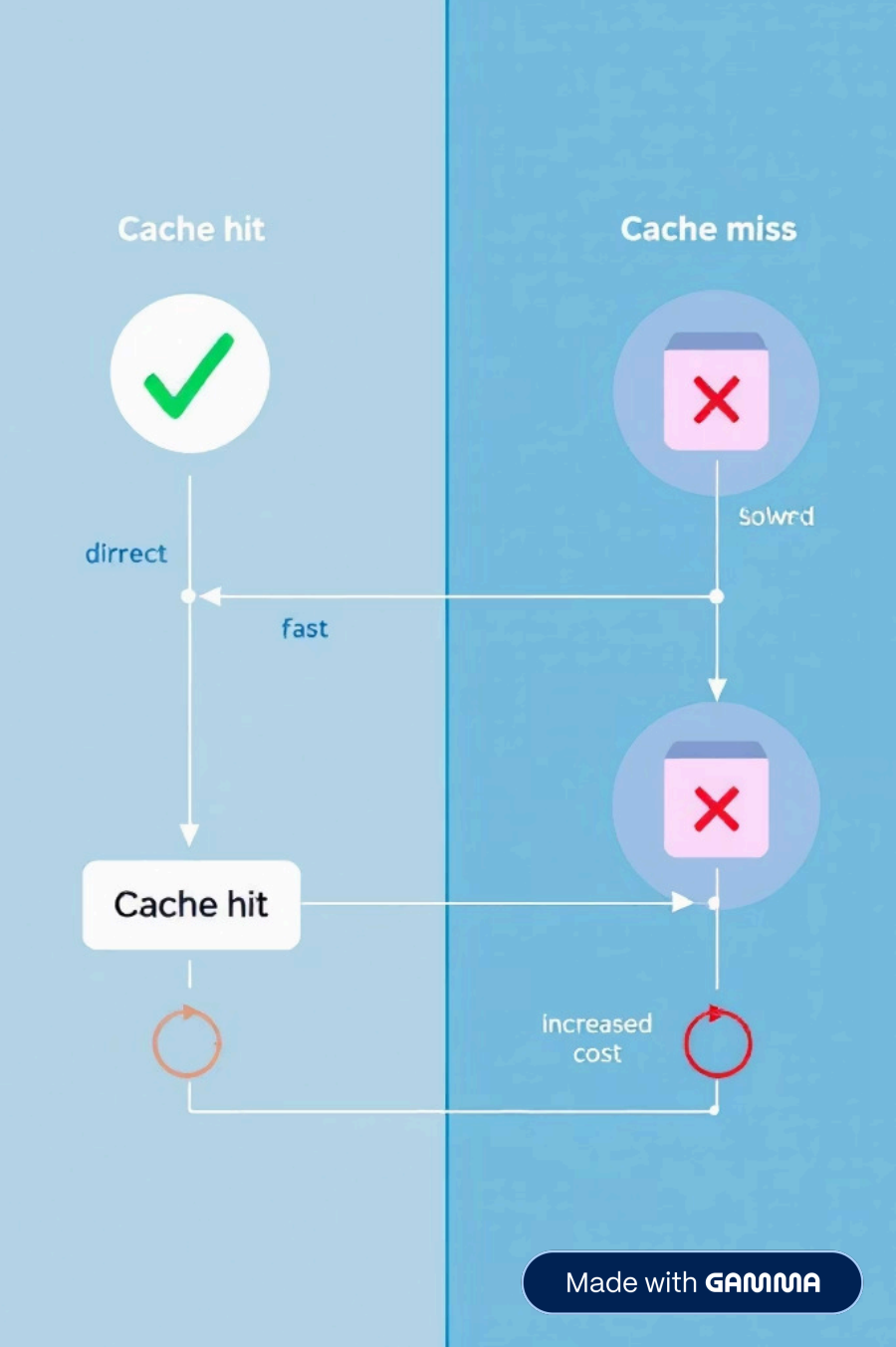
When you send a prompt, the provider stores computed key-value pairs representing the model's understanding. Subsequent requests with identical prefix retrieve cached states instead of recomputing them.

Cost impact

Cache hits cost approximately 10% of standard input tokens (90% discount). Cache writes cost 1.25-2x standard input tokens. For document Q&A with 50,000-token reference: first query costs $0.1875, subsequent queries cost $0.015 each.

When to apply

High-repeat intents, FAQ answering, multi-turn sessions with stable context, document Q&A applications, and batched document retrieval.



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Pricing for Image Inputs & Outputs

Image inputs are converted to token-equivalent counts based on resolution and detail level. For example, a typical 1024×1024 image is roughly equivalent to 700-1,300 tokens. Google Gemini charges 1,290 tokens for a 1024×1024 image, while OpenAI charges based on a detail parameter (low ~85 tokens, high scales with resolution). Image generation uses tiered pricing by resolution and quality.

- Image inputs: billed as token equivalents (preprocessing cost + vision model compute) based on resolution and detail.
- Image generation: per-image tiered cost by quality. OpenAI GPT Image costs $0.01 (low), $0.04 (medium), $0.17 (high) for square images. Google Imagen charges $30-120 per million output tokens depending on the model and resolution.

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API Pricing Comparison by Provider

High-level differentiation (estimates):

OpenAI (GPT series)

Wide model range from GPT-4o to GPT-4o-mini. Premium pricing for latest large models. Mixed billing for reasoning and multimodal features.

Anthropic Claude

Competitive rates for safety-focused workloads. Pricing often favorable for longer context lengths. Batch API offers 50% discount.

Google Gemini

Tight integration with cloud services. Pricing varies with multimodal and enterprise features. Approximately $1,290 tokens per 1024×1024 image.

xAI Grok

Positioned as lower-cost fast models for chat-heavy use cases. Automatic caching enabled for all requests. Check image/multimodal add-ons for complete pricing.

Note: negotiate enterprise discounts and reserved capacity for predictable high-volume usage.

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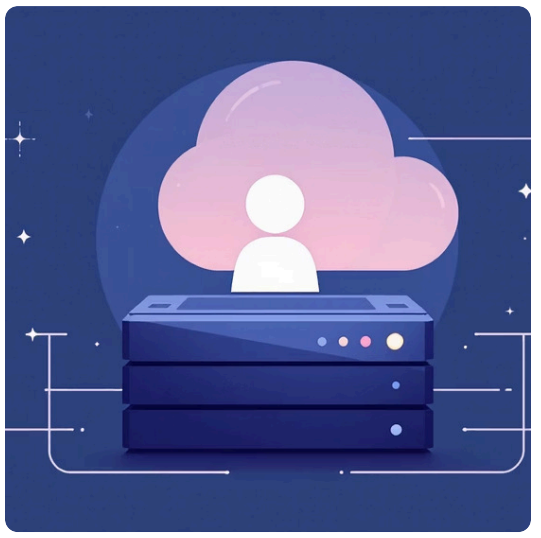
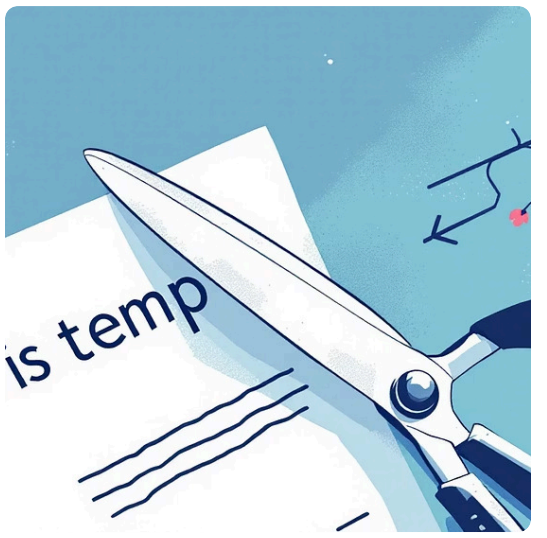
Optimization Strategies



Right-size models

Use smaller or distilled models for simple tasks like classification, simple Q&A, and routing decisions. Reserve largest models for complex reasoning, creative tasks, and high-stakes outputs. Route 70% of queries to cheap models, escalating only when needed.

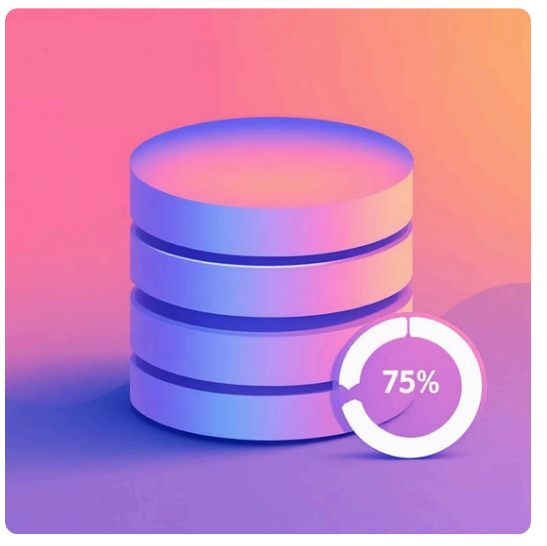


Trim prompts

Remove redundant instructions and verbose explanations. Use concise system prompts. Trim conversation history to include only relevant context. Use abbreviations or shorthand where the model can understand. Implement context summarization for long conversations.

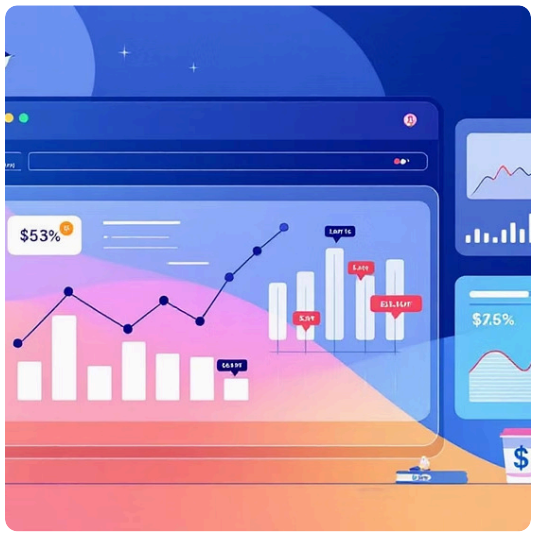
Enforce max outputs

Set appropriate max_tokens limits in API calls. Instruct the model to be concise in system prompt. Request structured outputs (JSON) instead of verbose prose. Use few-shot examples demonstrating desired response length.



Batch & cache

Combine similar requests into batches. Serve cached answers for repeated queries. Structure prompts with static content at beginning. Batch queries about same document together.



Monitor & budget

Implement per-environment budgets and alerting. Track token counts per feature and user. Add rate limiting and user quotas in app. Monitor usage patterns weekly and adjust approach.

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Real-World Cost Examples

01

Customer Support Chatbot

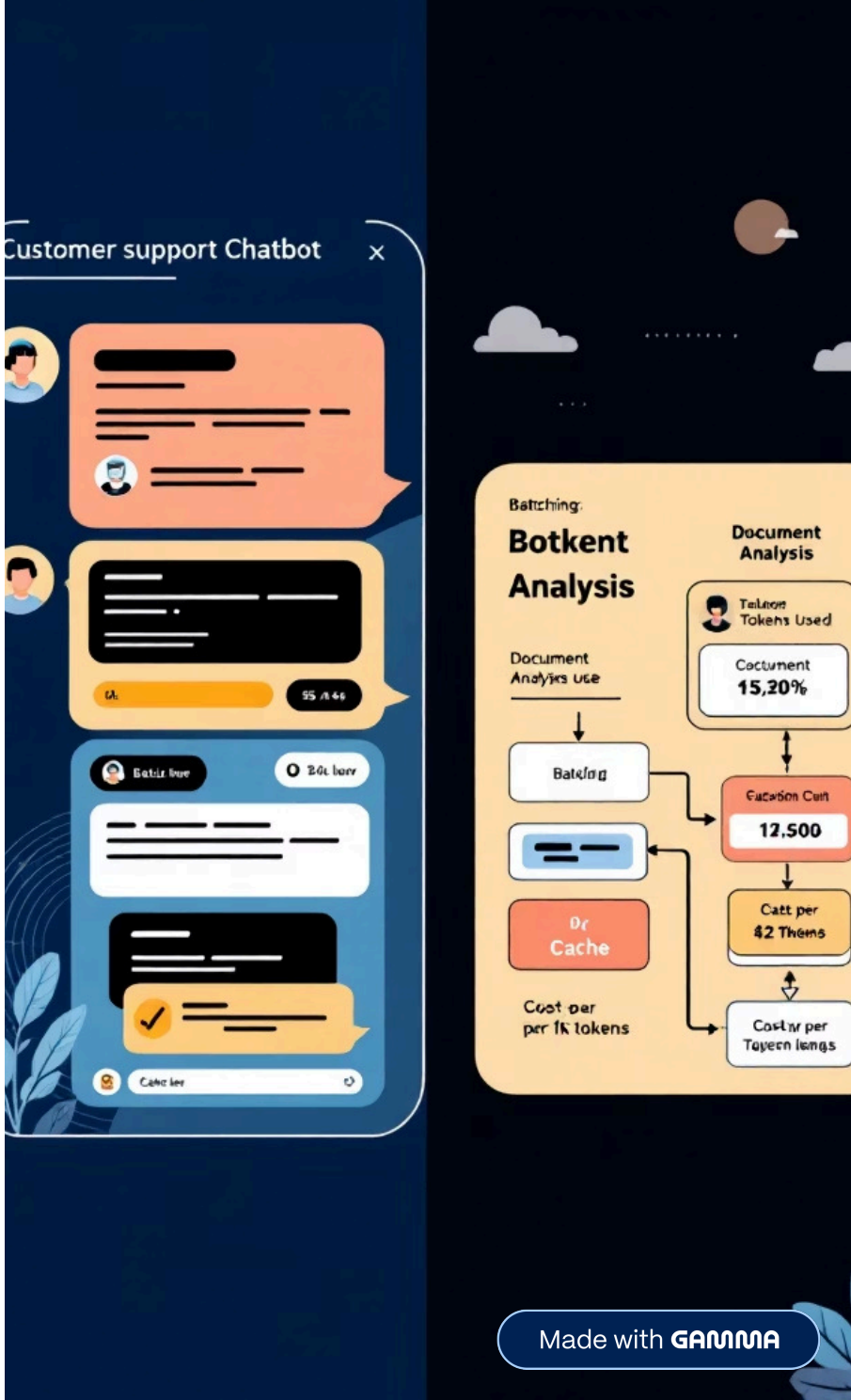
Assumptions: 50,000 questions/month, 500-token system prompt, 100-token average user query, 300-token average response. Input: 50,000 × 600 = 30M tokens. Output: 50,000 × 300 = 15M tokens. With Claude Sonnet 4.5 ($3/$15 per million): $90 + $225 = $315/month. Using smaller model for intent classification + caching templated replies can halve costs.

02

Document Analysis Pipeline

Process 1,000 contracts/month, 10,000 tokens each, 500-token summaries. Standard: 10M input tokens ($30) + 500K output tokens ($7.50) = $37.50/month. With Batch API (50% discount): $18.75/month. With Prompt Caching (80% hit rate on 2,000-token instructions): additional $4.32 savings.

Include sample cost tables in RFPs: per-1k-token estimates by model, plus expected monthly token volume to compare TCO across providers.



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Future Trends & Key Takeaways

- Price pressure will continue as competition increases — expect 20-30% annual price reductions as hardware efficiency improves, model architectures become more efficient (Mixture-of-Experts, speculative decoding), and competition from open-source models pressures pricing.
- Reasoning token economics emerging — thinking tokens can multiply output costs by 10-30x. Providers developing tiered pricing based on problem complexity.
- Model Optimizer and dynamic pricing — Google's Vertex AI Model Optimizer represents emerging paradigm where users specify desired outcomes (cost, quality, balance) rather than specific models.

Action items: run provider cost simulations with realistic token profiles, pilot caching + batching, negotiate enterprise terms for high-volume pipelines, benchmark new models quarterly as landscape evolves.

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